Let $CL = x$ and $LA = y$.

Then, $x + y = p$

In ΔABC and ΔLOC, we have

$\angle CAB = \angle CLO$ [each equal to $90°$]

$\angle C = \angle C$ [C is common]

$\Delta CAB \sim \Delta CLO$ [By AA similarity]

$$\frac{CA}{CL} = \frac{AB}{LO} \Rightarrow \frac{p}{x} = \frac{a}{h}$$

$$\text{so,} \quad x = \frac{ph}{a} \qquad \ldots(1)$$

In ΔALO and ΔACD, we have

$\angle ALO = \angle ACD$ [each equal to $90°$]

$\angle A = \angle A$ [A is common]

$\Delta ALO \sim \Delta ACD$ [by AA similarity]

$$\frac{AL}{AC} = \frac{OL}{DC} \Rightarrow \frac{y}{p} = \frac{h}{b} \text{ we get, } y = \frac{ph}{b} \qquad \ldots(2)$$

$$(1)+(2) \Rightarrow \quad x + y = \frac{ph}{a} + \frac{ph}{b}$$

$$p = ph\left(\frac{1}{a} + \frac{1}{b}\right) \qquad (\text{Since } x + y = p)$$

$$1 = h\left(\frac{a+b}{ab}\right)$$

Therefore, $h = \dfrac{ab}{a+b}$

Hence, the height of the intersection of the lines joining the top of each pole to the foot of the opposite pole is $\dfrac{ab}{a+b}$ metres.

Progress Check

1. All circles are _____ (congruent/ similar).
2. All squares are _____ (similar/ congruent).
3. Two triangles are similar, if their corresponding angles are _____ and their corresponding sides are _____.
4. (a) All similar triangles are congruent – True/False
   (b) All congruent triangles are similar – True/False.
5. Give two different examples of pair of non-similar figures?

Activity 1

Let us try to construct a line segment of length $\sqrt{2}$.

For this, we consider the following steps.

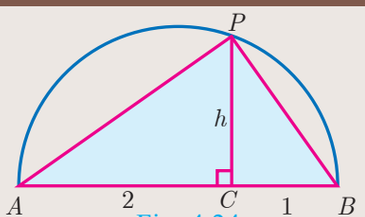


Fig. 4.24

Step1: Take a line segment of length 3 units. Call it as AB.

Step2: Take a point C on AB such that $AC = 2$, $CB = 1$.

Step3: Draw a semi-circle with AB as diameter as shown in the diagram

Step4: Take a point 'P' on the semi-circle such that CP is perpendicular to AB.

Step5: Join P to A and B. We will get two right triangles ACP and BCP.

Step6: Verify that the triangles ACP and BCP are similar.

Step7: Let $CP = h$ be the common altitude. Using similarity, find h.

Step8: What do you get upon finding h?

Repeating the same process, can you construct a line segment of lengths $\sqrt{3}$, $\sqrt{5}$, $\sqrt{8}$.

10th_Maths_Chapter 4_English.indd 168 12/27/2024 3:11:00 PM